

Experimental Design Report

Knowledge Distillation for Backward Compatibility:
Moving a Modern-Operator Model onto a Legacy Op Set

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Code and raw results: <https://github.com/austinchennn/kd-legacy-hw>

Scope of this document. A self-contained description of the experimental design: the research question, data, models, experimental groups, evaluation protocol, results (reported as obtained, including negative results), limitations, and where the conclusion stands.

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1 Research Question and Scope

Overall goal. New ML models increasingly depend on operators (GELU, LayerNorm, softmax attention) that older AI accelerators and edge DSPs, built around INT8 convolution pipelines, cannot execute natively. The question: *can a model restricted to a legacy operator set, trained with knowledge distillation (KD) from a modern-operator model, reproduce that model’s behaviour at acceptable accuracy and latency?*

Sub-questions.

- Q1. What does the op-set restriction cost in accuracy, size and latency?
- Q2. Does logit KD make the legacy student *behave more like* the teacher (prediction agreement), beyond accuracy?
- Q3. Does INT8 post-training quantization, the usual legacy-hardware format, preserve the result?

Explicit scope limitation.

Dimension	Coverage of this study
Task / data	Image classification, CIFAR-10 only
Teacher	One small ViT trained from scratch (no pre-training)
Student	One fixed VGG-style CNN architecture
KD method	Vanilla logit KD (Hinton et al., 2015) only
“Legacy hardware”	Emulated: op-set whitelist + INT8 PTQ + 1-thread CPU latency

The conclusion is therefore stated as “a feasibility study with an emulated legacy target”, not as a claim about any specific accelerator.

2 Dataset Design

- CIFAR-10: 50,000 train / 10,000 test, 32×32 RGB, 10 classes; standard per-channel normalization.
- Augmentation (train only): reflect-pad 4 + random 32×32 crop, horizontal flip ($p = 0.5$), executed on-device. *Implementation note:* the crop offset is sampled once per mini-batch (the flip is per-sample); this is weaker than per-sample cropping and applies identically to all groups.
- No validation split was carved out; no hyper-parameters were tuned on the test set (all settings were fixed before the runs; see §9).
- INT8 calibration: the first 2,048 training images.

3 Model Design

	Teacher (“new” model)	Student (“legacy-compatible”)
Architecture	ViT: patch 4 (64 tokens), dim 256, depth 6, 8 heads, MLP ratio 2, mean-pool head	VGG-style: $2 \times [\text{Conv3} \times 3\text{-BN-ReLU}]$ per stage, widths 32/64/128, MaxPool, GAP, Linear
Params	3.19M	0.29M
Operators	GELU, LayerNorm, softmax attention (<i>not</i> on legacy list)	Conv, BN, ReLU, MaxPool, Avg-Pool, Linear (all on legacy list)
Deployment	FP32	FP32 and INT8 (Conv-BN-ReLU fused, PTQ, qnnpack)

The legacy operator whitelist is `models.LEGACY_OPS`; compatibility is checked automatically over leaf modules.

4 Experimental Groups

Group	Model	Training signal	Epochs	Seeds
T	Teacher ViT	CE + label smoothing 0.1; AdamW 10^{-3} , wd 0.05	100	1
S	Student, scratch	CE; SGD-Nesterov 0.05, mom 0.9, wd 5×10^{-4}	40	3
K	Student, KD	$0.9 T^2 \text{KL}(p_t^T \ p_s^T) + 0.1 \text{CE}$, $T = 4$; same optimizer as S	40	3

All groups: batch 256, One-Cycle LR (15% warm-up). S and K differ *only* in the loss (same seeds, architecture, optimizer, schedule), so S is the control for K. Each student checkpoint is evaluated in FP32 and in INT8.

5 Evaluation Metric System

1. **Accuracy:** top-1 test accuracy.
2. **Behavioural compatibility:** agreement = fraction of test images where student and teacher predict the same class. Decomposed by teacher correctness: agreement on teacher-correct vs. teacher-wrong images.
3. **Legacy-hardware cost:** unsupported-op count; parameters; serialized size; batch-1 latency on **one CPU thread** (median of 300 runs after 30 warm-up; Apple M4 Max).
4. **Quantization robustness:** FP32 $\rightarrow$ INT8 change in accuracy and agreement.

Seeds: mean $\pm$ sample std over 3 seeds.

6 Results (reported as obtained)

6.1 Main table

Model	Acc (%)	Agree (%)	Params	Size	Lat. (ms)	Unsupp. ops
T: ViT, FP32	85.67	100	3.19M	12.8 MB	1.48	3
S: scratch, FP32	89.86 ± 0.13	83.63 ± 0.16	0.29M	1.17 MB	0.49	0
S: scratch, INT8	89.77 ± 0.14	83.53 ± 0.14	0.29M	0.30 MB	0.84	0
K: KD, FP32	89.25 ± 0.27	83.70 ± 0.31	0.29M	1.17 MB	0.48	0
K: KD, INT8	89.16 ± 0.30	83.62 ± 0.35	0.29M	0.30 MB	0.85	0

6.2 Agreement decomposition (FP32 students)

The teacher is correct on 85.67% of test images.

Quantity (% of relevant images)	S: scratch	K: KD
Agreement where teacher is <i>correct</i>	94.20 ± 0.15	93.93 ± 0.29
Agreement where teacher is <i>wrong</i>	20.45 ± 0.28	22.56 ± 0.53
Student right, teacher wrong (% of test set)	9.15 ± 0.01	8.78 ± 0.02
Student wrong, teacher right (% of test set)	4.97 ± 0.13	5.20 ± 0.25

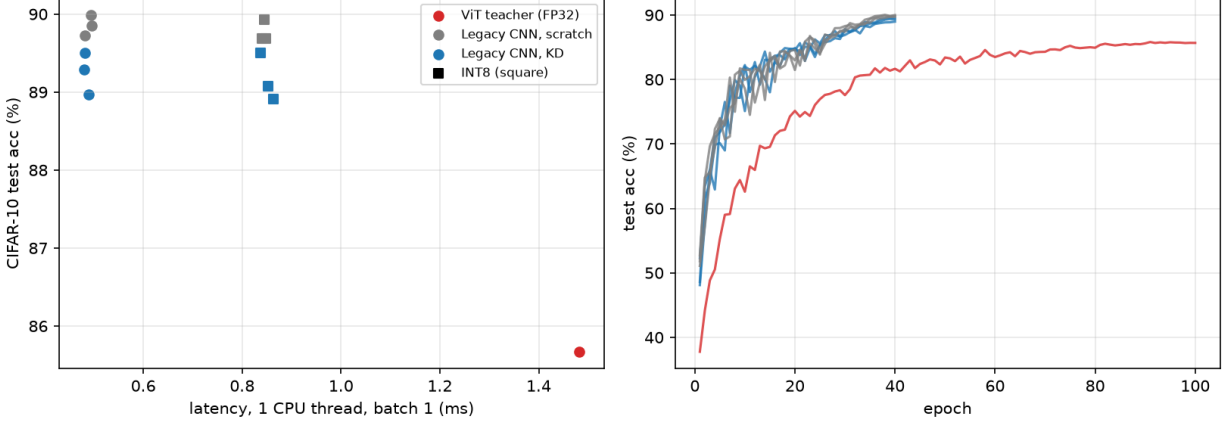


Figure 1: Left: accuracy vs. 1-thread batch-1 latency (circles FP32, squares INT8). Right: test accuracy per epoch (red teacher, gray scratch students, blue KD students).

6.3 Findings

- F1. **(Q1) The op-set restriction is cheap here.** The legacy student uses no unsupported operator, is $11\times$ smaller and $3.0\times$ faster (1 thread, FP32) than the teacher, and is *more* accurate (+4.2 pt).
- F2. **(Q2) Logit KD did not improve compatibility; it transferred the teacher’s errors.** Overall agreement is unchanged (83.70 vs. 83.63). The decomposition shows why: KD raises agreement on the teacher’s *mistakes* (20.45% $\rightarrow$ 22.56%, ≈ 5 pooled std) while slightly lowering it on the teacher’s correct predictions, so accuracy drops by 0.6 pt. Soft labels from a weaker, architecturally different teacher mainly carry its error modes.
- F3. **(Q3) INT8 PTQ preserves accuracy and agreement (≤ 0.1 pt change) and cuts size $3.9\times$.** However, **INT8 was slower than FP32 on this CPU** (0.84 vs. 0.49 ms): Apple Silicon’s FP32 path is highly optimized and qnnpack is not tuned for it. INT8 speed-ups are hardware-dependent, which is exactly why the target hardware should be part of the distillation objective.

7 Limitations and Validity

- **Weak teacher.** A from-scratch ViT on CIFAR-10 is below the CNN student; the realistic backward-compatibility setting (a stronger new model) is not covered. F2 may not hold with a stronger teacher.
- **Emulated hardware.** The op whitelist and 1-thread M4 latency are proxies, not measurements on a real legacy NPU/DSP.
- **Small scale.** One dataset, one student, one KD recipe ($T = 4$, $\alpha = 0.9$, not tuned), 3 seeds; one teacher run.
- **Augmentation detail.** Batch-shared crop offset (§2); identical across groups, so comparisons remain controlled.

8 Planned Next Experiments

Prio	Experiment	Purpose
P0	Stronger teacher (pre-trained ViT fine-tuned)	Test whether F2 is an artifact of the weak teacher
P0	KD sweep ($T \in \{1, 2, 4, 8\}$, α)	Separate “KD fails” from “KD untuned”
P1	Feature / attention-map distillation	Transfer intermediate representations across the attention→conv gap
P1	Correctness-gated / agreement-targeted loss	Match the teacher where it is right; do not copy its errors (motivated by F2)
P2	Real legacy target (e.g. Raspberry Pi, older mobile NPU)	Replace the emulated latency with measured latency; hardware-aware student search

9 Reproduction Notes

`PY=.venv/bin/python ./run_all.sh` (teacher 36.9 min; scratch student ≈ 2 min, KD student ≈ 5.7 min each, on MPS), then `python analyze.py` for §6.2. Raw per-epoch logs: `results/*.json`; benchmark: `results/bench.json`; decomposition: `results/agreement_breakdown.json`. PyTorch 2.14, torchvision 0.29, macOS, Apple M4 Max.

10 Where the Conclusion Stands

Established: on an emulated legacy target, a conv-only INT8 student can replace a ViT at lower cost with no accuracy loss, and vanilla logit KD across the attention→conv gap does not improve behavioural compatibility; instead, it measurably increases agreement on the teacher’s errors. Not established: whether this holds for strong teachers, other KD objectives, or real legacy hardware; these are the P0–P2 experiments above.